Critical Review:

Optimal Execution with Stochastic Delay

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Cycle of Conferences: Strategies and Actors of Portfolio Management

Course of Philippe Bergault

1 Introduction and context

In electronic markets, *latency* — the delay between the submission of an order and its notification by the exchange — constitutes a major risk for liquidity takers (Moallemi and Sağlam [8]). During this delay, the order book evolves and the targeted price may no longer be available. Despite the practical importance of this phenomenon in high-frequency trading, the literature on optimal execution assumed until now either zero latency (Almgren and Chriss [1], Guéant [6]) or deterministic latency (Bruder and Pham [2], Øksendal and Sulem [7]).

Cartea and Sánchez-Betancourt [3] propose the first optimal execution model with **stochastic latency**. A trader must liquidate an inventory of $\mathfrak{M}$ lots over a horizon $[0, T]$ by sending *Marketable Limit Orders* (MLOs) — orders for immediate execution equipped with a price limit protecting against adverse price moves. The problem is formulated as a **stochastic impulse control** problem [4] whose solution is characterised by a system of Hamilton–Jacobi–Bellman quasi-variational inequalities (HJBQVI). This work follows [5] on the cost of latency for fill rates.

2 Model summary

2.1 Order types and the central role of MLOs

The model relies on the distinction between **Market Orders** (MOs, no price limit, guaranteed execution but slippage incurred) and **Marketable Limit Orders** (MLOs, with a price limit protecting against adverse moves but risking being *missed*). MLOs represent 56.9% of EUR/USD volume on LMAX (Table 1), and 57.1% of them are missed — hence the importance of modelling this friction. From a financial perspective, a speculative MLO sent with latency can be interpreted as a **free short-lived option**: the trader captures the *price improvement* if the flicker is favourable (gain), and the order is simply cancelled otherwise (zero loss). Like any option, its value increases with the local volatility of flickers.

2.2 Price dynamics and flickers

The best bid price observed by the trader is modelled as

$$\hat{S}_t = S_t + F_t \,, \tag{1}$$

where S_t is the *fundamental* bid price and F_t is the *flicker* — a short-lived deviation caused by rapid arrivals and cancellations of orders in the book. Flickers only affect the model at MLO processing times.

Notification times and flicker sizes are modelled by a *marked point process* (MPP) $\mathcal{N} = (T_n, Z_n)_{n\geq 1}$, with i.i.d. interarrival times following an exponential distribution with parameter $\lambda > 0$. The flickers

Z_n are i.i.d. with mixture law:

$$\nu(\mathrm{d}z) = p_0\,\delta_{\{0\}}(\mathrm{d}z) + \underbrace{p_+\eta_+ e^{-\eta_+ z}\mathbf{1}_{\{z>0\}}\,\mathrm{d}z}_{\text{price improvement}} + \underbrace{p_-\eta_- e^{\eta_- z}\mathbf{1}_{\{z<0\}}\,\mathrm{d}z}_{\text{slippage}}\,, \tag{2}$$

with $p_0 + p_+ + p_- = 1$ and $\eta_+, \eta_- > 0$.

2.3 Stochastic latency and impulse control

Latency — the time between the submission of an order at time τ and its notification $\tilde{\tau}$ — is exponentially distributed with parameter λ (Lemma 3.1 of the paper). This memoryless property is crucial: it makes the system Markovian and allows the application of dynamic programming.

The trader's strategy is $\alpha = (\tau_i, \mathsf{l}_i)_{i\geq 1}$, where τ_i is the submission time of the i-th MLO and l_i its price limit. The key assumption is that at most one order can be *pending* (awaiting notification) in the exchange: $\tau_{i+1} \geq \tilde{\tau}_i > \tau_i$.

The system state $X_t^\alpha = (S_t^\alpha, Q_t^\alpha, C_t^\alpha)$ (fundamental price, inventory, cash) evolves according to the SDE:

$$X_t^\alpha = X_0 + \int_0^t \boldsymbol{b}(X_u^\alpha)\,\mathrm{d}u + \int_0^t \boldsymbol{\sigma}(X_u^\alpha)\,\mathrm{d}W_u + \sum_{\tilde{\tau}_i \leq t} \big(\Gamma(X^\alpha_{\tilde{\tau}_i^-}, \mathsf{F}_{\tilde{\tau}_i}, \mathsf{l}_i) - X^\alpha_{\tilde{\tau}_i^-}\big), \tag{3}$$

where $\boldsymbol{b}$ and $\boldsymbol{\sigma}$ (Lipschitz-continuous) are respectively the drift and volatility of the fundamental price (affecting only the first component of X), and the impulse operator $\Gamma : \mathbb{R}^3 \times \mathbb{R} \times L \to \mathbb{R}^3$ describes the transition at notification:

$$\Gamma(s, q, c, \mathsf{f}, \ell) = \big(s - \kappa f(\ell - s - \mathsf{f}),\ q - f(\ell - s - \mathsf{f}),\ c + (s + \mathsf{f})\, f(\ell - s - \mathsf{f})\big). \tag{4}$$

Here, f is a $\mathcal{C}^\infty$ approximation of the Heaviside step function $\tilde{H}(x) = \mathbf{1}_{\{x\leq 0\}}$, and $\kappa \geq 0$ is the permanent price impact parameter.

2.4 Performance criterion and value functions

The trader maximises:

$$\Pi(\alpha) = g(X_T^\alpha) + \int_0^T h(X_s^\alpha)\,\mathrm{d}s + \sum_{\tilde{\tau}_i \leq T} \mathfrak{C}(X_{\tilde{\tau}_i^-}, \mathsf{F}_{\tilde{\tau}_i}, \mathsf{l}_i)\,, \tag{5}$$

where $g(s, q, c) = c + q(s + \zeta - a(q-1))(1-\rho)$ is the terminal payoff ($\zeta = \int_{\mathbb{R}} z\,\nu(\mathrm{d}z)$ is the expected flicker, a the penalty for *walking the book*), $h(x) = -\phi q^2$ is the running inventory penalty (with $\phi \geq 0$ the *urgency* parameter), and $\mathfrak{C}(s, q, c, \mathsf{f}, \ell) = -\rho(s+\mathsf{f})\, f(\ell - s - \mathsf{f})$ represents transaction costs.

Denoting by J_0 and J_1 the expected value of this criterion over the remaining interval $[t, T]$ conditional on the state x (depending on whether there are respectively 0 or 1 pending orders), we define the value functions:

$$v_1(t, x, \ell) = \sup_{\alpha \in \mathcal{A}_{t,\ell}} J_1(t, x, \ell, \alpha) \qquad \text{(one pending order with limit } \ell), \tag{6}$$

$$v_0(t, x) = \sup_{\alpha \in \mathcal{A}_t} J_0(t, x, \alpha) \qquad \text{(no pending order)}. \tag{7}$$

2.5 HJBQVI system and viscosity solutions

The value functions satisfy the dynamic programming equation (Theorem A.5):

$$\min\left\{ -\frac{\partial v_0}{\partial t}(t, x) - \mathcal{L}v_0(t, x) - h(x),\ \ v_0(t, x) - \sup_{\ell \in L} v_1(t, x, \ell) \right\} = 0 \quad \text{on } \mathcal{D}_0\,, \tag{8}$$

and

$$\frac{\partial v_1}{\partial t} + \mathcal{L}v_1 + h(x) + \lambda\,\mathbb{E}\big[v_0\big(t, \Gamma(x, Z, \ell)\big) + \mathfrak{C}(x, Z, \ell)\big] - \lambda\, v_1 = 0 \quad \text{on } \mathcal{D}_1\,, \tag{9}$$

with $\mathcal{L} = b(s)\frac{\partial}{\partial s} + \frac{1}{2}\sigma^2(s)\frac{\partial^2}{\partial s^2}$ the infinitesimal generator of the price, and $\mathcal{D}_0$, $\mathcal{D}_1$ the state domains depending on whether there are 0 or 1 pending orders.

The interpretation is as follows: equation (8) expresses that the trader either waits or sends an MLO with the optimal price limit; equation (9) describes the evolution while awaiting notification (λ = arrival rate). To solve this system numerically, the authors exploit the ansatz $v_0(t, s, q, c) = c + h_0(t, s, q)$, which eliminates the cash variable and reduces the problem from 4D to 3D, making the finite difference scheme tractable.

Theorem A.9 establishes uniqueness of the viscosity solution of (8) under the growth condition:

$$\sup_{(t,x)\in\mathcal{D}_0} \frac{|v_0(t,x)|}{1+|x|} < \infty\,, \tag{10}$$

which is satisfied when b and σ are constant.

3 Empirical results

The authors use proprietary data from LMAX Exchange (EUR/USD, September 2019 – February 2020) to estimate parameters and test the strategy. The estimated parameters for the 9:00am–9:10am window are (Table 4): $\hat{\sigma} = 2.1\times10^{-4}$, $\hat{p}_+ = 0.05$, $\hat{p}_- = 0.08$, $1/\hat{\eta}_+ = 1.87\times10^{-5}$, $1/\hat{\eta}_- = 2.66\times10^{-5}$, $\hat{a} = 1.80 \times 10^{-5}$.

The RLOS (*Random-Latency Optimal Strategy*) is compared to four benchmarks:

1. **DLOS**: optimal strategy with deterministic latency;
2. **ZLOS**: optimal strategy with zero latency;
3. **TWAP**: MOs at regular intervals;
4. **ENOW**: a single MO at time 0.

Main result. For a patient trader ($\phi = 0$) with an expected latency of 30 ms, RLOS outperforms TWAP by 3.80 \$/€M and ENOW by 2.57 \$/€M (Table 5). These amounts are comparable to transaction costs (c = 3 \$/€M), making them economically significant. RLOS and DLOS have statistically identical performances, suggesting that the exact distribution of latency matters little as long as the expectation is correct.

Patient vs impatient traders. The urgency parameter ϕ controls the strategy's behaviour. When $\phi = 0$, the trader uses the entire window to send ∼195 MLOs, of which ∼97% are speculative (Table 6). When $\phi = 10^{-3}$, only ∼10 non-speculative MLOs are sent to liquidate quickly (Table 7).

Around announcements. The outperformance is multiplied by 2 to 10 during news events (Table 10): for instance, 36.31 \$/€M vs ZLOS in the minute following the announcement, compared to 3.40 in normal periods. This is explained by the increase in the probability and size of *price improvements* ($\hat{p}_+ = 0.056$, $1/\hat{\eta}_+ = 2.52 \times 10^{-5}$ around news vs $\hat{p}_+ = 0.042$, $1/\hat{\eta}_+ = 1.92 \times 10^{-5}$ in calm periods, Table 11).

4 Critiques

4.1 Mathematical critiques

Exponential latency distribution. The central assumption of the model is that latency follows an exponential distribution with parameter λ. The *memoryless* property ($\mathbb{P}[\tilde{\tau} - \tau > t + s \mid \tilde{\tau} - \tau > s] = e^{-\lambda t}$) makes the problem Markovian, but it is empirically questionable: in practice, network latency follows a more concentrated distribution (log-normal, Gamma) and the notification probability should increase with waiting time, not remain constant.

Sigmoid approximation and numerical convergence. The impulse operator (4) uses a $\mathcal{C}^\infty$ approximation f^ε of $\tilde{H}(x) = \mathbf{1}_{\{x \le 0\}}$, necessary for the theory of viscosity solutions, but the error $\|f^\varepsilon - \tilde{H}\|$ and its impact on the value function are not quantified. More fundamentally, the authors acknowledge (p. 41) that the convergence of their finite difference scheme to the viscosity solution remains an "*open problem*", which weakens the rigour of the numerical validation.

Growth condition for uniqueness. Condition (10), essential for the uniqueness Theorem A.9, is only satisfied when $b(s) \equiv b$ and $\sigma(s) \equiv \sigma$ are constant. For more realistic price dynamics (state-dependent drift or volatility), uniqueness of the viscosity solution is no longer guaranteed.

Independence of flickers. The flickers (Z_n) are assumed i.i.d., which is a strong simplification. In practice, flickers exhibit *clustering* (activity begets activity), violating the independence assumption. The natural tool would be a **Hawkes process** whose self-exciting intensity incorporates the history of events, but this would cause the dimensionality of the HJBQVI system to explode (λ would become a state variable).

4.2 Model critiques

Restriction to a single pending order. The assumption $\tau_{i+1} \ge \tilde{\tau}_i$ (at most one pending MLO) is a major simplification. In practice, a fast trader with 30 ms latency and a 6 s horizon can send dozens of orders before receiving the first notification. Relaxing this assumption would expand the state space from $\{0,1\}$ to $\{0,1,\ldots,m\}$ pending orders, with m coupled value functions.

Theory/practice gap on price impact ($\kappa = 0$). The theoretical model incorporates permanent price impact via the parameter κ in the impulse operator (4). However, the empirical validation imposes $\kappa = 0$ because the backtest is performed on a *historical* order book: the fictitious RLOS orders cannot trigger reactions from market makers. The measured outperformance is therefore obtained on a "passive" market, which likely overestimates real gains. For an inventory of $\mathfrak{M} = 10$ lots (5 M EUR), this bias remains moderate; for larger volumes, the gap between theory ($\kappa > 0$) and validation ($\kappa = 0$) would be significant. Moreover, the theoretical model assumes unit orders that do not consume liquidity beyond the *best price*, but the backtest allows RLOS to "*walk down the LOB*" to fill remaining lots — an inconsistency between assumptions and empirical implementation.

Absence of strategic interaction and limited scope. The model assumes that other participants do not react to the trader's activity; in reality, *market makers* dynamically adjust their quotes. Furthermore, the horizon $T = 6\,\text{s}$ and $\mathfrak{M} = 10$ lots are specific to high-frequency FX, limiting generalisation.

5 Numerical example

To illustrate the model's mechanisms, we propose a simplified Monte Carlo simulation (Python code provided in a separate document). The model used is: flickers following the law (2) with the parameters from Table 4 ($p_0 = 0.87$, $p_+ = 0.05$, $p_- = 0.08$, $1/\eta_+ = 1.87\times10^{-5}$, $1/\eta_- = 2.66\times10^{-5}$), exponential latency with variable mean, and a heuristic for the speculative/non-speculative choice. We compare RLOS to TWAP and ENOW over 50,000 trajectories.

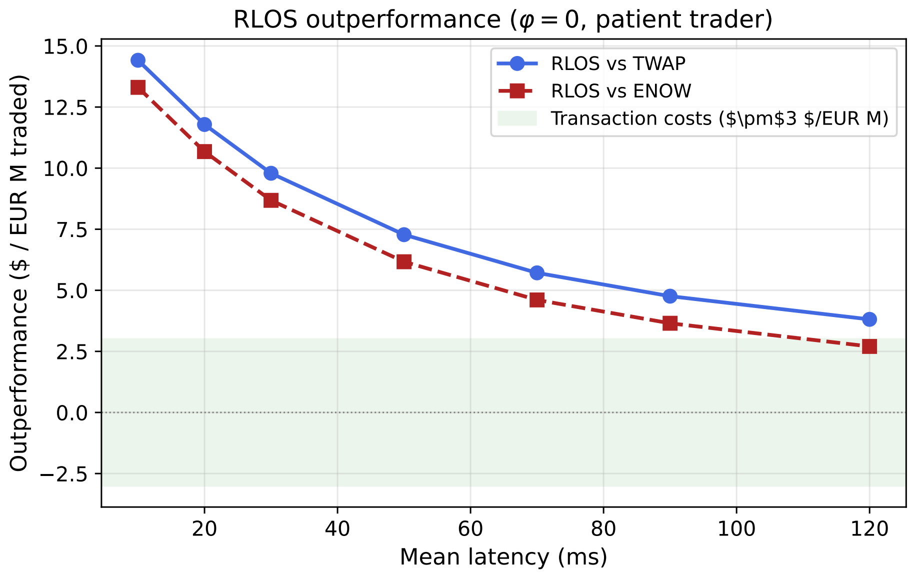

Figure 1: Outperformance of the simplified RLOS strategy relative to TWAP and ENOW as a function of mean latency. Outperformance decreases with latency as the trader has fewer opportunities for speculative MLOs.

Figure 1 qualitatively confirms the paper's results: RLOS outperformance decreases with mean latency. At 30 ms, the trader makes on average ∼105 MLO attempts, of which 95% are speculative (compared to ∼195 attempts and 97% speculative in Table 6). The fill rate of speculative MLOs is approximately 5%, consistent with $p_+ = 0.05$. We also note that the average TWAP gain is negative ($-1.11\,\$/€\text{M}$) due to the asymmetry of flickers ($p_- > p_+$, and $1/\eta_- > 1/\eta_+$). This asymmetry reflects *adverse selection*: facing informed flow, the book thins out before order processing, causing slippage more frequently than *price improvement*. Orders without *price protection* therefore bear a structural cost.

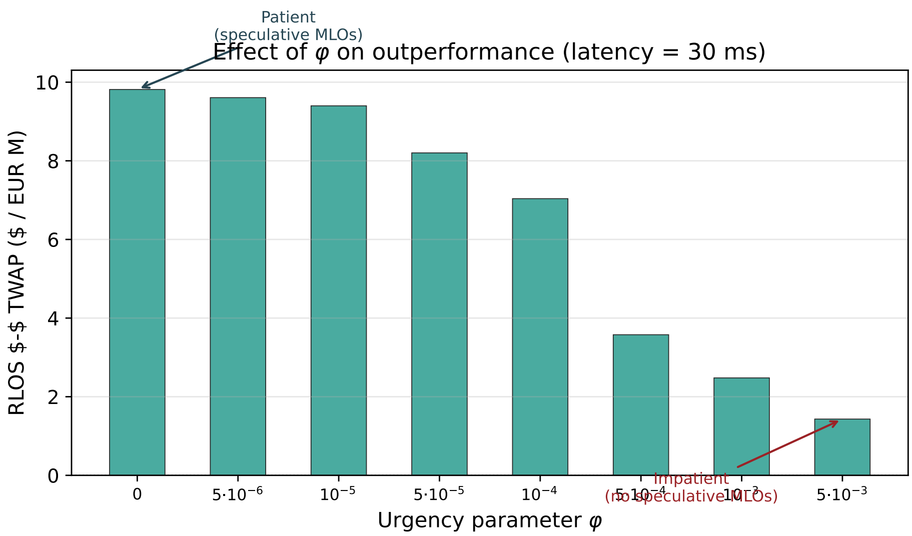

Figure 2: Effect of the urgency parameter φ on RLOS outperformance relative to TWAP (mean latency $= 30\,\text{ms}$). A patient trader ($\varphi \approx 0$) exploits speculative MLOs; an impatient trader (φ large) behaves like TWAP.

Figure 2 illustrates the patient/impatient dichotomy. For $\phi = 0$, 95% of MLOs are speculative and the outperformance reaches $+9.81\,\$/€\text{M}$. For $\phi = 5 \times 10^{-3}$, the speculative ratio drops to 8% and the outperformance falls to $+1.43\,\$/€\text{M}$: the trader, eager to liquidate, uses only normal MLOs and approaches TWAP (cf. Tables 6–7 of the paper).

Impact of the latency distribution. To test the critique on the exponential assumption (Section 4.1), we replace the exponential distribution with a Gamma(k, μ/k) distribution with the same mean μ but variance μ^2/k (coefficient of variation $1/\sqrt{k}$). Figure 3 compares the cases $k = 1$ (exponential), $k = 5$ and $k = 20$ (highly concentrated latency).

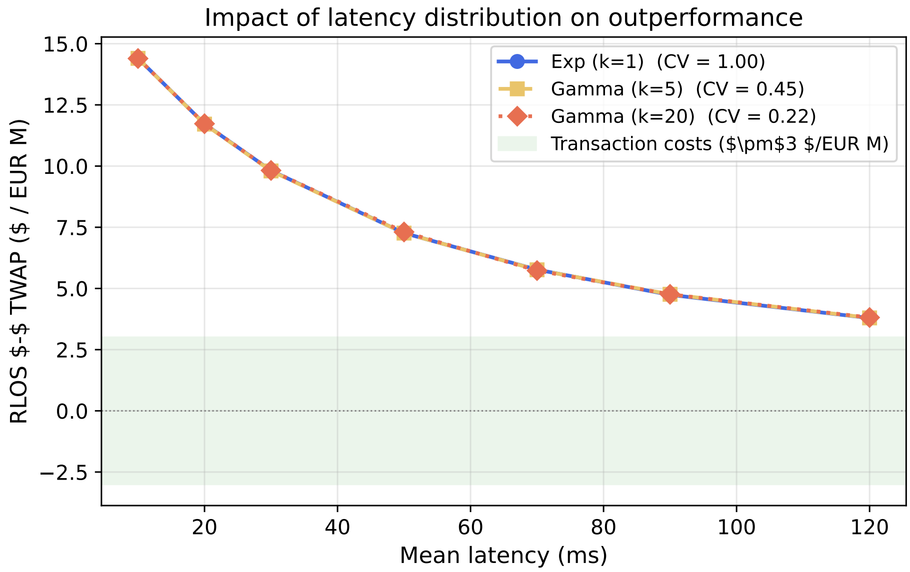

Figure 3: RLOS vs TWAP outperformance for three latency distributions with the same mean: exponential ($k = 1$, CV = 1), Gamma($k = 5$, CV = 0.45), Gamma($k = 20$, CV = 0.22). The curves overlap almost perfectly.

Notable result: the three curves are nearly identical. However, this reflects a property of our heuristic, **mathematically insensitive** to the shape of the distribution (only the mean matters via the expected number of remaining attempts), and not a robustness of the model. A truly optimal strategy under a Gamma distribution would exploit the increasing hazard rate $h(t) = f(t)/(1-F(t))$ to adjust MLO timing — an open problem.

Remark. These results are qualitative (simplified heuristic, not the full HJBQVI system), but the orders of magnitude (∼4–14 \$/€M) are of the same order as those of the paper (3–5 \$/€M), the gap being explained by our simplified heuristic.

6 Extensions and future directions

More realistic latency distributions. Replacing the exponential with a Gamma or log-normal distribution within the full HJBQVI framework poses a major challenge: the problem loses the Markov property and requires *state augmentation* (adding the time elapsed since submission). The optimal strategy could then exploit the increasing hazard rate of these distributions.

Multiple pending orders. Allowing $m > 1$ simultaneous pending orders is the most natural extension: $m + 1$ coupled value functions would replace the current two-function system. Bruder and Pham [2] treat the deterministic case; the stochastic extension remains open and relevant given the ∼195 attempts in 6 s (Table 6).

Numerical resolution via Deep Learning. These extensions aggravate the *curse of dimensionality*; Deep Learning methods (*Deep Galerkin Method*, *Deep BSDE*) constitute a promising avenue. Other extensions include endogenous impact (propagator) and multi-asset execution with differentiated latencies.

7 Conclusion

The paper by Cartea and Sánchez-Betancourt opens a new research direction by formulating optimal execution with stochastic latency as an impulse control problem. The validation on LMAX data is convincing (outperformance of 3–5 \$/€M). The main limitations — exponential latency, single pending order, zero-impact backtest — call for ambitious extensions (Hawkes, multiple orders, Deep Learning) that would bring the model closer to high-frequency trading practice.

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